

Day 24: Data Refresh & On-Premises Gateways

[DEF] Definition to Remember

The On-Premises Data Gateway acts as a secure cryptographic bridge that enables Power BI Service cloud datasets to query and refresh data from local on-premises databases and network file shares.

1. Gateway Types & Architecture

* Standard Mode (Enterprise Gateway): Installed on a dedicated server; supports multiple users, scheduled refreshes, and DirectQuery live connections across enterprise SQL and Oracle servers.

[Power BI Service Cloud]
(Encrypted HTTPS Port 443)

[On-Premises Gateway Service] (Behind Corporate Firewall)
(Local Network Query)

[Local SQL Server / Excel Files]

2. Scheduled Refresh Limits

* Power BI Pro: Up to 8 scheduled refreshes per day per semantic model.

3. Incremental Refresh

* Instead of reloading 10 years of historical data on every daily refresh, Incremental Refresh:

[MUST-WRITE] Must-Master Skills & Interview Points

1. How the On-Premises Gateway works without opening inbound corporate firewall ports.
2. Diagnosing common refresh failures (expired credentials, network timeouts, M query folding breaks).
3. Configuring Incremental Refresh with RangeStart and RangeEnd.

[RECALL] Quick Recall

Enterprise Gateway (Secure Firewall Bridge) + Scheduled Refresh (8x Pro / 48x Premium) + Incremental Refresh